HARSH TOMAR

AI/ML Engineer Generative AI RAG

Builds GenAI systems across retrieval, tool use, model adaptation, evaluation, and deployment.

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SUMMARY

AI/ML engineer focused on **Generative AI, RAG, agentic systems, and model implementation**. Built **OncoRAG**, an evidence-grounded oncology agent; **Continuum**, an 11-microservice LLMOps evaluation platform; and **QuantaAI**, a LangGraph search-routing system with 94% routing-classifier accuracy on a hand-labelled evaluation set. Hands-on with Python, PyTorch, LangChain, LangGraph, vector databases, and LLM APIs.

EXPERIENCE

AI Engineer Intern **May 2025 – Jul 2025**
i3 Digital Health *Remote*

- Built and deployed a **RAG pipeline for researcher profiling**, aggregating and cleaning data from PubMed, Google Scholar, and ResearchGate via multi-source APIs and scraping; **LangChain** retrieval over vector embeddings drove contextual collaborator recommendations
- Owned the full build-to-deploy loop: iterated chunking strategies, embedding models, and retrieval prompts with healthcare SMEs; when their judgment on trust diverged from my retrieval metrics, rebuilt the system around source attribution and a refuse-when-unsure rule

PROJECTS

OncoRAG: Evidence-Grounded Oncology Agent **2026**

Weaviate, hybrid search, Claude, FastAPI | **GitHub**

- Built an agent over CIViC, ClinicalTrials.gov, and openFDA with tool-based retrieval, citation validation, and a full search trace for auditability
- Evaluated a hand-verified set and adversarial cases: 28/28 grounding-clean, 25/28 citation-ID matches, and 10/10 red-team cases; surfaced citation mismatches rather than hiding them

Continuum: LLMOps Evaluation & Observability Platform **2026**

AWS ECS Fargate, Apache Kafka, FastAPI, React/TypeScript, Terraform, PEFT, MLflow, Prometheus | **GitHub**

- **Problem:** detect quality degradation in evaluated LLM traces and design a controlled remediation path
- **Approach: 11-microservice event-driven platform** on **AWS ECS Fargate** with an **Apache Kafka** trace pipeline; multi-dimensional quality scoring (Claude LLM-judge + ROUGE/BLEU + semantic similarity); **Statistical Process Control** (Western Electric rules) for drift detection; designed a controlled **LoRA** fine-tuning path with A/B canary deployment; full **Terraform IaC** and GitHub Actions CI/CD
- **Result:** verified end-to-end trace flow (Kafka to scoring to PostgreSQL) with **166 Prometheus metric series**, an authenticated REST API, a React dashboard, and 10/10 passing E2E tests

QuantaAI: LangGraph Search-Routing Chatbot **2025**

LangGraph, GPT-4, Tavily Search, FastAPI, Next.js 14, TypeScript | **GitHub**

- **Problem:** chat assistant that decides on its own when a query needs live web search vs. direct generation
- **Approach: 4-node LangGraph state machine** (Classifier → Search → Generate → Response) with a GPT-4 routing classifier; **Tavily** web-search integration with relevance ranking; SSE streaming via FastAPI to a Next.js 14 frontend; sliding 10-message context window
- **Result: 94% routing-classifier accuracy** on a hand-labelled evaluation set; transparent multi-stage UX exposing classification, search, and generation phases

VLMverse: PaLiGemma Vision-Language Model **2025**

PyTorch, SigLIP, Gemma 2B, RoPE, KV-Cache, Grouped-Query Attention, RMSNorm | **GitHub**

- **Problem:** internalise modern VLM architecture by reimplementing one end to end

- **Approach:** reimplemented **PaLiGemma** in pure PyTorch following the paper and reference walkthroughs: SigLIP encoder (16×16 patches to 196 tokens), **Gemma-2B** decoder, **RoPE** positional embeddings, **KV-Cache** for autoregressive inference, grouped-query attention, RMSNorm
- **Result:** working end-to-end inference; deep working knowledge of attention, positional encodings, and cross-modal projection

ClinicalQwen: Medical Q&A Fine-Tuning on Qwen 2.5 1.5B **2026**

PyTorch, Qwen 2.5 1.5B, QLoRA, 4-bit NF4, PEFT, Hugging Face | **GitHub**

- **Problem:** adapt a general-purpose LLM to answer clinical and medical queries accurately
- **Approach: QLoRA** fine-tuning of **Qwen 2.5 1.5B** on **5,000 medical Q&A samples** (lavita/medical-qa-datasets); 4-bit NF4 quantisation; PEFT with LoRA rank-8 adapters; trained on Google Colab T4
- **Result:** domain-adapted model with ∼1% trainable parameters (9.2M / 1,552M); 86% keyword coverage, perplexity 3.14; deployed on Hugging Face Hub

Additional Work

- **issue-resolver-agent** *(2026)*: autonomous GitHub issue fixer; SOUL.md behavioural spec + RULES.md hard constraints; escalation rules stop the agent when ambiguous or high-risk **[GitHub]**
- **Transformers-CV** (**173 commits**, *2024–2026*): PyTorch reimplementations of ViT, DETR, Swin, SAM, Flamingo VLM, DDPM, JEPA, VAE/VQ-VAE from scratch **[GitHub]**
- **RAGCraft** *(2024)*: a hub for RAG patterns, including advanced chunking, hybrid search, and re-ranking across Weaviate, Pinecone, FAISS, and Chroma **[GitHub]**
- **Inference Engineering Study** (forked, in progress): working through KV-Cache, speculative decoding, quantisation (FP8/INT8/INT4), vLLM, TensorRT-LLM **[GitHub]**

WRITING

- **Statistical Process Control for LLM Quality Monitoring**: Continuum's drift-detection engine, covering WECO rules, EWMA, and three honest gaps in the implementation **[Read]**
- **Deconstructing Autoresearch**: a technical teardown of Andrej Karpathy's autoresearch repo, covering attention patterns, scaling behaviour, and optimiser internals **[Read]**

OPEN SOURCE CONTRIBUTIONS

3 merged PRs to upstream projects:

- BBoxMaskPose #12: Docker support for the ICCV 2025 paper repo
- multimodal-agents-course #23: fixed a broken Kubrick UI build
- hive #6849: refactored a deprecated storage API

TECHNICAL SKILLS

LLM & GenAI: LangChain, LangGraph, LlamaIndex, RAG, prompt engineering, agent orchestration, OpenAI / Anthropic / Gemini APIs, Hugging Face Transformers
Fine-Tuning & Architecture: LoRA, QLoRA, PEFT, 4-bit NF4 quantisation, attention mechanisms, RoPE, KV-Cache, grouped-query attention, RMSNorm
Vector & Retrieval: Pinecone, Chroma, FAISS, embedding models, semantic search, chunking strategies, re-ranking
Deployment & LLMOps: FastAPI, Docker, Kubernetes, AWS (ECS Fargate, EKS, EC2, S3, ECR), Apache Kafka, Terraform, MLflow, Prometheus, Grafana, GitHub Actions CI/CD, Hugging Face Spaces
Foundations: PyTorch (primary), TensorFlow, NumPy, Pandas, transformers, autoregressive decoding |
Languages: Python, TypeScript, Bash

EDUCATION

B.Tech in Artificial Intelligence & Data Science **Nov 2022 – June 2026**
Lakshmi Narain College of Technology, Bhopal CGPA: 7.37/10
Coursework: NLP, Deep Learning, Reinforcement Learning, Linear Algebra, Probability & Statistics, Neural Networks